EU4 Innovation
Kosovo - eID

eID Platform API

This project is funded
by the European Union

A project implemented by
EVOLUXER

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1. Summary

The primary aim of this document is to present a comprehensive and detailed description of the various APIs offered by the eID platform. These APIs are designed to facilitate seamless and efficient integration with third-party applications that require their functionality. By providing in-depth information and guidance, this document seeks to ensure that developers and stakeholders can easily understand and implement the APIs to enhance the interoperability and capabilities of their systems.

2. eID platform API

The following chapter highlights the APIs made available by the eID platform.

The APIs are split into three groups:

- **CMS management**
  - Enable RA
  - List RAs
  - Disable RA
  - Revoke certificate
  - Cancel request
  - List requests
  - List users
  - Delete User
- **End user Life cycle management**
  - Login
  - Check password
  - Check password availability
  - Generate OTP
  - Verify OTP
  - Get documents
  - Change PIN
  - Change PUK
  - Unlock PIN
  - Recovery ERC
  - Revoke certificates
  - Ext Recovery username
  - Ext recovery ERC
  - Ext recovery password
  - Logout
- **Signature and Timestaping**
  - Echo
  - Send OTP
  - Sign
  - Sign with timestamp
  - Timestamp

2.1. Base URL

Environment	Hostname
TEST	eid-cms.kren-ks.eu
PRODUCTION	eid-cms.rks-gov.net

2.2. Signature and Timestamping base URL

In pre-production the base URL to be used to invoke the API below is as follows:

`http://eid.kren-ks.eu`

In production, instead, the base URL to be used to invoke the API below is as follows:

`https://eid-signcloud.rks-gov.net`

Depending on whether you want to apply a signature or a timestamp, only the path changes, which for the signature is **`/signbox`** while for the timestamp is **`/tsa`**

2.3. API user token permissions

In this section we will give evidence of the permissions associated with each type of token that can be created on the eID platform.

The process of generating a token is described in the eID platform user manual.

The possible roles to associate with the creation of an application token are as follows:

1. Api Backend User
2. Registration Authority Officer
3. Manager
4. Auditor
5. Administrator

Below is a table containing the roles/permissions mapping

Role	Permissions
Api Backend User	• Enable RA • Disable RA • Cancel request • Revoke certificate
Registration Authority Officer	• Add citizen request • Add business request • View request • Change request

Role	Permissions
	• Approve request • Download contract • Verify contact • Cancel request • Add document file • View document file • Change document file • Delete document file • View RA • Revoke certificate
Manager	The same permissions as RAOs + these one: • Add RA • Delete RA • Enable RA • Disable RA • Add user • View user • Change user • Delete user • View log
Auditor	• View request • Download contract • View RA • View user • View API user • View document file • View email template • View document template • View log
Administrator	The same permissions as RAO and Manager + these one: • Resend credentials • Add Api user • Change Api user • Delete Api user • Add document file • Change document file • Delete document file • Add document template • Change document template • Delete document template • Add email template • Change email template

Role	Permissions
	• Delete email template

3. CMS management API

3.1. Registration Authority

3.1.1. Enable Registration Authority

Description: the following API allows you to enable a Registration Authority previously created.

Scope: Admin

URL: `https://{host} /api/cms/registrationauthorities/{ID_RA}/enable/`

Method: PUT

Header parameters:

- **Authorization:** Token {YOUR_API_TOKEN}

Parameters in request body:

- **No parameters**

Sample response in case of success:

- **HTML Code:** 200 OK
- **Content:**

```
{
    "enabled": true
}
```

Sample response in case of error (ID_RA not present):

- **HTML Code:** 404 Not found
- **Content:**

```
{
    "detail": “Not found.”
}
```

3.1.2. List of Registration Authorities

Description: the following API allows you to get all the information related to an RA, included the Operators associated.
The API return a result for both RA enabled and disabled.

Scope: RAO

URL: `https://{host} /api/cms/registrationauthorities/?name={RA_NAME}`

Method: `GET`

Header parameters:

- **Authorization:** Token {YOUR_API_TOKEN}

Parameters in request body:

- **No parameters**

Sample response in case of success:

- **HTML Code:** 200 OK
- **Content:**


```
{
    "count": 1,
    "next": null,
    "previous": null,
    "results": [
        {
            "pk": 24,
            "name": "RA6",
            "parent": 3,
            "email": "fdo@bit4id.com",
            "phone_number": "+38300000000",
            "postal_code": "",
            "address": "Test6",
            "city": 162043,
            "province": null,
            "country": "XK",
            "operators": [
                {
                    "pk": 70,
                    "username": "fabio.donofrio@ra6",
```

```
                        "first_name": "Fabio",
                        "last_name": "D'Onofrio",
                        "is_active": true,
                        "fiscal_code": "2004813157",
                        "groups": [
                            "Manager"
                        ]
                    }
                ],
                "vat_number": "123556781",
                "manager_fiscal_code": "2004813157",
                "manager_first_name": "Fabio",
                "manager_last_name": "D'Onofrio"
            }
        ]
    }
```


Sample response in case of error (RA_NAME not present):

- **HTML Code:** 200 OK
- **Content:**


```
{
    "count": 0,
    "next": null,
    "previous": null,
    "results": []
}
```


3.1.3. Disable Registration Authority

Description: the following API allows you to disable a Registration Authority previously created.

Scope: Admin

URL: `https://{host} /api/cms/registrationauthorities/{ID_RA}/disable/`

Method: PUT

Header parameters:

- **Authorization:** Token {YOUR_API_TOKEN}

Parameters in request body:

- **No parameters**

Sample response in case of success:

- **HTML Code:** 200 OK
- **Content:**

```
{
    "enabled": false
}
```

Sample response in case of error (ID_RA not present):

- **HTML Code:** 404 Not found
- **Content:**

```
{
    "detail": “Not found.”
}
```

3.2. Requests

3.2.1. Revoke certificates

Description: the following API allows to revoke a user’s certificate.

Scope: RAO

URL: `https://{host} /api/cms/requests/{ID_REQUEST}/revoke_certificate/`

Method: PUT

Header parameters:

- **Authorization:** Token {YOUR_API_TOKEN}

Parameters in request body:

- **erc:** ERC code received via email post request approval
- **reason**: reason for requesting revocation

Request body

```
{
    "erc": "string",
    "reason": "string"
```

```
}
```

Sample response in case of success:

- **HTML Code:** 200 OK
- **Content:**

```
{
    "enabled": true
}
```

Sample response in case of error (ID_RA not present):

- **HTML Code:** 404 Not found
- **Content:**

```
{
    "detail": “Not found.”
}
```

3.2.2. Cancel request

Description: the following API allows to cancel a request.

Scope: RAO

URL: `https://{host} /api/cms/requests/{ID_REQUEST}/cancel/`

Method: PUT

Header parameters:

- **Authorization:** Token {YOUR_API_TOKEN}

Parameters in request body:

- **No parameters**

Sample response in case of success:

- **HTML Code:** 200 OK
- **Content:**

```
{
    "status": ok
}
```

Sample response in case of error (when the request status is not CREATED, VERIFYING_CONTACTS, VERIFIED_CONTACTS, CANCELED):

- **HTML Code:** 403 Forbidden
- **Content:**

```
{
    "detail": "You do not have permission to perform this action."
}
```

Sample response in case of error (ID_RA not present):

- **HTML Code:** 404 Not found
- **Content:**

```
{
    "detail": "Not found."
}
```

3.2.3. List request

Description: given a personal number and the id of an RA, the API returns information about all requests related to the entered personal number and belonging to the entered RA.

Scope: RAO

URL:
`https://{host}/api/cms/requests/?fiscal_code={PERSONAL_NUMBER}&registration_authority={RA_ID}`

Method: GET

Header parameters:

- **Authorization:** Token {YOUR_API_TOKEN}

Parameters in request body:

- **No parameters**

Sample response in case of success:

- **HTML Code:** 200 OK
- **Content:**

```
{
    "count": 2,
    "next": null,
```

```
"previous": null,
"results": [
    {
        "pk": 89,
        "type": "CITIZEN",
        "status": "ISSUED",
        "registration_authority": 3,
        "registering_user": "superadmin",
        "approving_user": "superadmin",
        "first_name": "Ahmet",
        "last_name": "Jetullahu",
        "sex": "M",
        "birth_date": "1972-01-12",
        "birth_country": "XK",
        "birth_city": 162047,
        "fiscal_code": "2004813157",
        "email": "fdo@bit4id.com",
        "phone_number": "+38349406474",
        "country": "XK",
        "city": 161998,
        "address_name": "BREGU I DIELLIT, OB-3, RR.3, HY-34, NR.1",
        "postal_code": "",
        "id_document_type": "ID",
        "id_document_number": "ID1558124",
        "id_document_issuer_type": "MUP",
        "id_document_issue_date": "2020-01-10",
        "id_document_expiration_date": "2030-01-09",
        "company": null,
        "documents": [
            {
                "pk": 215,
                "name": "ID-Front-89.PDF",
                "status": "NOT_ARCHIVED",
                "type": "ID",
                "format": "PDF",
                "side": "FR",
                "archived_date": null,
                "signature_status": "NOT_SIGNED"
```

```
                },
                {
                    "pk": 217,
                    "name": "Contract-89.PDF",
                    "status": "NOT_ARCHIVED",
                    "type": "CC",
                    "format": "PDF",
                    "side": "FR",
                    "archived_date": null,
                    "signature_status": "SIGNED"
                },
                {
                    "pk": 216,
                    "name": "ID-Back-89.PDF",
                    "status": "NOT_ARCHIVED",
                    "type": "ID",
                    "format": "PDF",
                    "side": "BK",
                    "archived_date": null,
                    "signature_status": "NOT_SIGNED"
                }
            ],
            "certificate_set": [
                {
                    "status": "REVOKED",
                    "serial_number":
"5c39ae6f564085dcb068757e69b7b0b07f44db8b",
                    "subject": "2.5.4.46=89, CN=Ahmet Jetullahu,
2.5.4.5=PNOXK-2004813157, 2.5.4.42=Ahmet, 2.5.4.4=Jetullahu, C=XK",
                    "issuer": "CN=Republic of Kosovo",
                    "valid_from": "2024-11-14T10:00:39+01:00",
                    "valid_to": "2026-11-14T10:00:38+01:00",
                    "profile": "Remote Signature",
                    "data": "{BASE64 CERTIFICATE}
                }
            ],
            "profile": "qualified",
            "identification_type": "DEVISU",
```

```
        "token_identifier": "ahmet.jetullahu#89"
    },
    {
        "pk": 64,
        "type": "COMPANY",
        "status": "CANCELED",
        "registration_authority": 3,
        "registering_user": "fabio.donofrio@central-ra",
        "approving_user": null,
        "first_name": "Test",
        "last_name": "Test",
        "sex": "M",
        "birth_date": "1980-01-01",
        "birth_country": "XK",
        "birth_city": 162047,
        "fiscal_code": "2004813157",
        "email": "fdo@bit4id.com",
        "phone_number": "+383 49 406 474",
        "country": "XK",
        "city": 161998,
        "address_name": null,
        "postal_code": "",
        "id_document_type": "ID",
        "id_document_number": "2004813156",
        "id_document_issuer_type": "MPB",
        "id_document_issue_date": "2018-12-11",
        "id_document_expiration_date": "2028-12-11",
        "company": {
            "vat_number": "810000711",
            "name": "Bujar Olluri B.I.",
            "postal_code": "",
            "address_type": "",
            "address_name": "MARSHI UÇK",
            "address_number": "",
            "country": "XK",
            "province": 297,
            "city": 161998,
            "application": "Bujar Olluri B.I."
```

```
    },
    "documents": [
        {
            "pk": 125,
            "name": "ID-Front-64.PDF",
            "status": "NOT_ARCHIVED",
            "type": "ID",
            "format": "PDF",
            "side": "FR",
            "archived_date": null,
            "signature_status": "NOT_SIGNED"
        },
        {
            "pk": 126,
            "name": "ID-Back-64.PDF",
            "status": "NOT_ARCHIVED",
            "type": "ID",
            "format": "PDF",
            "side": "BK",
            "archived_date": null,
            "signature_status": "NOT_SIGNED"
        },
        {
            "pk": 127,
            "name": "Company-registration-64.PDF",
            "status": "NOT_ARCHIVED",
            "type": "CR",
            "format": "PDF",
            "side": "FR",
            "archived_date": null,
            "signature_status": "NOT_SIGNED"
        },
        {
            "pk": 128,
            "name": "Company-authorization-64.PDF",
            "status": "NOT_ARCHIVED",
            "type": "CA",
            "format": "PDF",
```

```
                    "side": "FR",
                    "archived_date": null,
                    "signature_status": "NOT_SIGNED"
                }
            ],
            "certificate_set": [],
            "profile": "qualified_seal",
            "identification_type": "DEVISU",
            "token_identifier": null
        }
    ]
}
```


Sample response in case of error (PERSONAL NUMBER not VALID):

- **HTML Code:** 200 OK
- **Content:**


```
{
    "count": 0,
    "next": null,
    "previous": null,
    "results": []
}
```


Sample response in case of error (RA_ID not VALID):

- **HTML Code:** 400 Bad request
- **Content:**


```
{
    "registration_authority": [
        "Select a valid choice. That choice is not one of the available
choices."
    ]
}
```


Sample response in case of error (RA_ID on which there are no requests related to the personal number entered):

- **HTML Code:** 200 OK
- **Content:**


```
{
    "count": 0,
    "next": null,
    "previous": null,
```

```
        "results": []
    }
```

3.3. User

3.3.1. List User

Description: given a username and the RA's ID, the API returns information about the user.

Scope: RAO

URL: `https://{host}/api/cms/users/?username={USERNAME}&registration_authority={RA_ID}`

Method: `GET`

Header parameters:

- **Authorization:** Token {YOUR_API_TOKEN}

Parameters in request body:

- **No parameters**

Sample response in case of success:

- **HTML Code:** 200 OK
- **Content:**

```
{
    "count": 1,
    "next": null,
    "previous": null,
    "results": [
        {
            "pk": 50,
            "username": "fabio.donofrio@central-ra",
            "first_name": "Fabio",
            "last_name": "D'Onofrio",
            "email": "fdo@bit4id.com",
            "groups": [
                "Registration Authority Officer"
            ],
```

```
                "is_active": true,
                "fiscal_code": "0000000000",
                "registration_authority": 3
            }
        ]
    }
```


Sample response in case of error (RA_ID not VALID):

- **HTML Code:** 400 Bad request
- **Content:**


```
{
    "registration_authority": [
        "Select a valid choice. That choice is not one of the available
choices."
    ]
}
```


Sample response in case of error (RA_ID on which the username inserted does not exists):

- **HTML Code:** 200 OK
- **Content:**


```
{
    "count": 0,
    "next": null,
    "previous": null,
    "results": []
}
```


Sample response in case of error (USERNAME not valid):

- **HTML Code:** 200 OK
- **Content:**


```
{
    "count": 0,
    "next": null,
    "previous": null,
    "results": []
}
```


3.3.2. Delete User

Description: the following API allows to delete a user.

Scope: Admin

URL: `https://{host} /api/cms/users/{USER_ID}/delete/`

Method: `DELETE`

Header parameters:

- **Authorization:** Token {YOUR_API_TOKEN}

Parameters in request body:

- **No parameters**

Sample response in case of success:

- **HTML Code:** 200 OK
- **Content:**

```
{
    "deleted": true
}
```

Sample response in case of error (USER_ID already deleted or does not exists):

- **HTML Code:** 404 Not found
- **Content:**

```
{
    "detail": "Not found."
}
```

4. End user life-cycle management API

The Kosovo eID platform API provides endpoints to manage user authentication, password verification, credential creation, and other administrative tasks for an electronic identification system. The API uses token-based authentication.

Pre-requisite: before using the API below, you must have a valid signature certificate issued by the eID platform.

4.1. Login

Description: the following API allows you to login the Self-Service portal of the eID platform.

URL: `https://{host}/api/lcm/login/`

Method: POST

Header parameters:

- **Authorization:** Token {YOUR_API_TOKEN}

Parameters in request body:

- **username**: the username that the user receives via email downstream of the RAO's approval of the request
- **password**: the password that the user receives in the encrypted PDF attached to the post-approval email of the request

Request body:


```
{
    "username": "string",
    "password": "string"
}
```


Sample response in case of success:

- **HTML Code:** 200 OK
- **Content:**


```
{
    "status": "ok",
    "session":
"13991feeb2d4ff82ee7f155104806c97f42d703c72ac611d0c33a7ebf29cab6f",
    "request": {
        "pk": 89,
        "type": "CITIZEN",
        "status": "ISSUED",
        "registration_authority": 3,
        "registering_user": "*******",
        "approving_user": "*******",
        "first_name": "*****",
        "last_name": "*******",
        "sex": "M",
        "birth_date": "*******",
        "birth_country": "Kosovo (XK)",
        "birth_city": "Dranoc",
        "fiscal_code": "********",
```

"email": "********",
"phone_number": **********",
"country": "Kosovo (XK)",
"city": "Ferizaj",
"address_name": "BREGU I DIELLIT, OB-3, RR.3, HY-34, NR.1",
"postal_code": "",
"id_document_type": "ID",
"id_document_number": "*******",
"id_document_issuer_type": "MUP",
"id_document_issue_date": "*******",
"id_document_expiration_date": "*******",
"company": **null**,
"documents": [
    {
        "pk": 215,
        "name": "ID-Front-89.PDF",
        "status": "NOT_ARCHIVED",
        "type": "ID",
        "format": "PDF",
        "side": "FR",
        "archived_date": **null**,
        "signature_status": "NOT_SIGNED"
    },
    {
        "pk": 217,
        "name": "Contract-89.PDF",
        "status": "NOT_ARCHIVED",
        "type": "CC",
        "format": "PDF",
        "side": "FR",
        "archived_date": **null**,
        "signature_status": "SIGNED"
    },
    {
        "pk": 216,
        "name": "ID-Back-89.PDF",
        "status": "NOT_ARCHIVED",
        "type": "ID",

```
                "format": "PDF",
                "side": "BK",
                "archived_date": null,
                "signature_status": "NOT_SIGNED"
            }
        ],
        "certificate_set": [
            {
                "status": "VALID",
                "serial_number": "5c39ae6f564085dcb068757e69b7b0b07f44db8b",
                "subject": "2.5.4.46=89, CN=*******, 2.5.4.5=PNOXK-2004813157,
2.5.4.42=*******, 2.5.4.4=*******, C=XK",
                "issuer": "CN=Republic of Kosovo",
                "valid_from": "2024-11-14T10:00:39+01:00",
                "valid_to": "2026-11-14T10:00:38+01:00",
                "profile": "Remote Signature",
                "data": "{BASE64 CERTIFICATE}"
            }
        ],
        "profile": "Remote Signature",
        "identification_type": "DEVISU",
        "token_identifier": "{USERNAME}",
        "is_revokable": true,
        "is_suspendable": false,
        "is_resumable": false,
        "is_renewable": false
    },
    "token_info": {
        "qualified": true,
        "puk_counter": 3,
        "pin_counter": 3,
        "identifier": "{USERNAME}",
        "webunlock": false,
        "label": "{USERNAME}_Label"
    },
    "password_changed": false
}
```

NOTE: the session value returned by the login API must be used in all subsequent APIs where it is requested.

4.2. Check password

Description: the following API allows a check of the password used to access the Self-service portal.

URL: `https://{host}/api/lcm/check_password/`

Method: POST

Header parameters:

- **Authorization:** Token {YOUR_API_TOKEN}

Parameters in request body:

- **session**: session id returned from the login API response (par. 4.1)
- **password**: the password that the user receives in the encrypted PDF attached to the post-approval email of the request

Request body:

```
{
    "session": "string",
    "password": "string"
}
```

Sample response in case of success:

- **HTML Code:** 200 OK
- **Content:**

```
{
    "status": "ok",
}
```

4.3. Check password validity

Description: the following API enables a check of the password used to access the Self-service portal of the eID platform.

URL: `https://{host}/api/lcm/check_password_validity/`

Method: POST

Header parameters:

- **Authorization:** Token {YOUR_API_TOKEN}

Parameters in request body:

- **session**: session id returned from the login API response (par. 4.1)
- **password**: the password that the user receives in the encrypted PDF attached to the post-approval email of the request

Request body:

```
{
    "session": "string",
    "password": "string"
}
```

Sample response in case of success:

- **HTML Code:** 200 OK
- **Content:**

```
{
    "status": "ok",
}
```

4.4. Generate OTP

Description: the following API allows you to proceed with generating and sending an OTP code from a seed associated with the user's identity.

URL: `https://{host}/api/lcm/generate_otp/`

Method: POST

Header parameters:

- **Authorization:** Token {YOUR_API_TOKEN}

Parameters in request body:

- **session**: session id returned from the login API response (par. 4.1)
- **uses**: represents the number of uses of the OTP

Request body:

```
{
    "session": "string",
    "uses": integer
```

```
}
```

Sample response in case of success:

- **HTML Code:** 200 OK
- **Content:**

```
{
    "status": "ok",
    "phone_number": "**********",
    "wait": "180000",
}
```

4.5. Verify OTP

Description: the following API allows you to proceed with the verification of the correctness of the OTP inserted

URL: `https://{host}/api/lcm/verify_otp/`

Method: POST

Header parameters:

- **Authorization:** Token {YOUR_API_TOKEN}

Parameters in request body:

- **session**: session id returned from the login API response (par. 4.1)
- **otp**: represents the generated OTP of the generate_otp API (par. 4.4)

Request body:

```
{
    "session": "string",
    "otp": "string"
}
```

Sample response in case of success:

- **HTML Code:** 200 OK
- **Content:**

```
{
    "status": "ok",
```

```
}
```

4.6. Get documents

Description: the following API allows you to obtain all documents related to a given request.

URL: `https://{host}/api/lcm/documents/`

Method: POST

Header parameters:

- **Authorization:** Token {YOUR_API_TOKEN}

Parameters in request body:

- **session**: session id returned from the login API response (par. 4.1)

Request body:

```
{
    "session": "string",
}
```

Sample response in case of success:

- **HTML Code:** 200 OK
- **Content:**

```
[
    {
        "pk": 217,
        "request": 89,
        "name": "Contract-89.PDF",
        "status": "NOT_ARCHIVED",
        "type": "CC",
        "format": "PDF",
        "side": "FR",
        "notes": null,
        "archived_date": null,
        "data": "{BASE64 DOC}",
        "signature_status": "SIGNED"
```

```
        }
    ]
```

4.7. Change PIN

Description: the following API allows you to proceed with the PIN change.

URL: `https://{host}/api/lcm/change_pin/`

Method: POST

Header parameters:

- **Authorization:** Token {YOUR_API_TOKEN}

Parameters in request body:

- **session**: session id returned from the login API response (par. 4.1)
- **pin:** current PIN
- **new_pin**: new PIN

Request body:

```
{
    "session": "string",
    "pin": "string",
    "new_pin": "string"
}
```

Sample response in case of success:

- **HTML Code:** 200 OK
- **Content:**

```
{
    "status": "ok"
}
```

4.8. Change PUK

Description: the following API allows you to proceed with the modification of the PUK.

URL: `https://{host} /api/lcm/change_puk/`

Method: POST

Header parameters:

- **Authorization:** Token {YOUR_API_TOKEN}

Parameters in request body:

- **session**: session id returned from the login API response (par. 4.1)
- **puk:** current PUK
- **new_puk**: new PUK

Request body:

```
{
    "session": "string",
    "puk": "string",
    "new_puk": "string"
}
```

Sample response in case of success:

- **HTML Code:** 200 OK
- **Content:**

```
{
    "status": "ok"
}
```

4.9. Unlock PIN

Description: the following API allows you to proceed with PIN unlocking.

URL: `https://{host}/api/lcm/unlock_pin/`

Method: POST

Header parameters:

- **Authorization:** Token {YOUR_API_TOKEN}

Parameters in request body:

- **session**: session id returned from the login API response (par. 4.1)
- **puk:** current PUK

- **new_pin**: new PIN

Request body:

```
{
    "session": "string",
    "puk": "string",
    "new_pin": "string"
}
```

Sample response in case of success:

- **HTML Code:** 200 OK
- **Content:**

```
{
    "status": "ok"
}
```

4.10. Recovery ERC

Description: the following API allows you to proceed with ERC code retrieval via email or phone number.

URL: `https://{host}/api/lcm/recovery_erc/`

Method: POST

Header parameters:

- **Authorization:** Token {YOUR_API_TOKEN}

Parameters in request body:

- **session**: session id returned from the login API response (par. 4.1)
- **notifier:** corresponds to the method used to notify the user of the ERC; it can have EMAIL or SMS value

Request body:

```
{
    "session": "string",
    "notifier": "string"
}
```

Sample response in case of success:

- **HTML Code:** 200 OK
- **Content:**

```
{
    "status": "ok"
}
```

4.11. Revoke certificates

Description: the following API allows you to proceed with the revocation of a certificate. The certificate being revoked must be in the "Valid" state; if the API is called for a certificate that has already been revoked, it will respond with an appropriate message.

For all other states that are not "Valid" or "Revoked", the API will respond with a 403 Forbidden.

URL: `https://{host}/api/lcm/revoke_certificates/`

Method: POST

Header parameters:

- **Authorization:** Token {YOUR_API_TOKEN}

Parameters in request body:

- **session**: session id returned from the login API response (par. 4.1)
- **erc:** ERC code received via email post request approval
- **reason**: reason for requesting revocation

Request body:

```
{
    "session": "string",
    "erc": "string"
    "reason": "string"
}
```

Sample response in case of success:

- **HTML Code:** 200 OK
- **Content:**

```
{
```

```
    "status": "ok"
}
```

4.12. Ext recovery username

Description: the following API allows you to proceed with username retrieval outside the login session.

URL: `https://{host}/api/lcm/ext_recovery_username/`

Method: POST

Header parameters:

- **Authorization:** Token {YOUR_API_TOKEN}

Parameters in request body:

- **fiscal_code**: corresponds to the user's personal number
- **email**: user email address

Request body:

```
{
    "fiscal_code": "string",
    "email": "string"
}
```

Sample response in case of success:

- **HTML Code:** 200 OK
- **Content:**

```
{
    "status": "ok"
}
```

4.13. Ext recovery ERC

Description: the following API enables the retrieval of the ERC code outside the login session.

URL: `https://{host}/api/lcm/ext_recovery_erc/`

Method: POST

Header parameters:

- **Authorization:** Token {YOUR_API_TOKEN}

Parameters in request body:

- **fiscal_code**: corresponds to the user's personal number
- **username**: the username that the user receives via email downstream of the RAO's approval of the request

Request body:

```
{
    "fiscal_code": "string",
    "username": "string"
}
```

Sample response in case of success:

- **HTML Code:** 200 OK
- **Content:**

```
{
    "status": "ok"
}
```

4.14. Ext recovery password

Description: the following API allows you to proceed with password recovery outside the login session.

URL: `https://{host}/api/lcm/ext_recovery_password/`

Method: POST

Header parameters:

- **Authorization:** Token {YOUR_API_TOKEN}

Parameters in request body:

- **username**: the username that the user receives via email downstream of the RAO's approval of the request
- **erc**: emergency code

Request body:

```
{
```

```
    "username": "string"
    "erc": "string"
}
```

Sample response in case of success:

- **HTML Code:** 200 OK
- **Content:**

```
{
    "status": "ok"
}
```

4.15. Logout

Description: the following API allows you to logout of the Self Service portal.

URL: `https://{host}/api/lcm/logout/`

Method: POST

Header parameters:

- **Authorization:** Token {YOUR_API_TOKEN}

Parameters in request body:

- **session**: session id returned from the login API response (par. 4.1)

Request body:

```
{
    "session": "string"
}
```

Sample response in case of success:

- **HTML Code:** 200 OK
- **Content:**

```
{
    "status": "ok"
}
```

5. Signature and Timestamping API

The APIs described in this chapter provide a set of interfaces for integrating file signing and timestamping capabilities into third-party applications

5.1. Echo

Description: replies back what it receives to check the service availability.

URL: `https://{host}/signbox/api/v1/echo?message=test`

Method: `GET`

Header parameters:

- **No parameters**

Parameters in request body:

- **No parameters**

Sample response in case of success:

- **HTML Code:** 200 OK
- **Content:** {STRING}

Sample response in case of error:

- **HTML Code:** 401 Unauthorized
- **Content:**

```
{
  "error_code": 0,
  "detail": "string"
}
```

- **HTML Code:** 422 Unprocessable Entity
- **Content:**

```
{
  "error_code": 0,
  "detail": "string"
}
```

5.2. Send OTP

Description: generates and send an OTP via SMS for the specified digital identity.

URL: `https://{host}/signbox/api/v1/otp`

Method: POST

Header parameters:

- **No parameters**

Parameters in request body:

- **username**: required, represent the digital identity username
  - type: {STRING}
- **password**: required, represent the digital identity password
  - type: {STRING}
- **uses**: optional, this parameter represents the max uses of the OTP. If not specified, the API can be used for massive signature; the important thing in this case is that no more than 3 minutes elapses between each signature.
  - type: {INTEGER}

Sample response in case of success:

- **HTML Code:** 200 OK
- **Content:** {}

Sample response in case of error:

- **HTML Code:** 422 Unprocessable Entity
- **Content:**

```
{
  "error_code": 0,
  "detail": "string"
}
```

5.3. Sign

Description: signs the submitted file.

URL: `https://{host}/signbox/api/v1/sign`

Method: POST

Header parameters:

- **No parameters**

Parameters in request body:

- **file**: required, represent the file to be signed
    - type: {STRING}
- **username**: required, represent the digital identity username
    - type: {STRING}
- **password**: required, represent the digital identity password
    - type: {STRING}
- **pin**: required, represent the digital identity PIN
    - type: {STRING}
- **format**: Types of supported signature profiles:
    - PADES for PDF files
        - type: {STRING}
    - XADES for XML files
        - type: {STRING}
    - CADES for other files
        - type: {STRING}
- **otp**: required, represent the OTP code previously sent to the user via SMS.
    - type: {STRING}
- **level**: optional, represent the signature level to address the need to protect the validity of the signature in time.
    - Values:
        - BASIC (the lowest and simplest version containing at least a signature value, a reference to or a copy of the signing certificate as a signed attribute)
        - T (A timestamp regarding the time of signing is added to protect against repudiation.)
    - type: {STRING}

Sample response in case of success:

- **HTML Code:** 200 OK
- **Content:** file signed

Sample response in case of error:

- **HTML Code:** 401 Unauthorized
- **Content:**

```
{
  "error_code": 0,
  "detail": "string"
}
```

5.4. Sign with timestamp

Description: apply a signature and a timestamp to the submitted file.

URL: `https://{host}/signbox/api/v1/sign`

Method: `POST`

Header parameters:

- **No parameters**

Parameters in request body:

- **file**: required, represent the file to be timestamped
  - type: {STRING}
- **username**: required, represent the digital identity username
  - type: {STRING}
- **password**: required, represent the digital identity password
  - type: {STRING}
- **pin**: required, represent the digital identity PIN
  - type: {STRING}
- **otp**: required, represent the OTP code previously sent to the user via SMS.
  - type: {STRING}
- **format**: optional, supported timestamp formats
  - supported format:
    - PDF
    - P7M
  - type: {STRING}

- **level**: optional, represent the signature level to address the need to protect the validity of the signature in time.
    - supported value:
        - BASIC (the lowest and simplest version containing at least a signature value, a reference to or a copy of the signing certificate as a signed attribute)
        - T (a timestamp regarding the time of signing is added to protect against repudiation.)
- **tsa_url**: required, represent the timestamp service URL required for signatures with level T
    - type: {STRING}

Sample response in case of success:

- **HTML Code:** 200 OK
- **Content:** file signed and timestamped

Sample response in case of error:

- **HTML Code:** 401 Unauthorized
- **Content:**

```
{
  "error_code": 0,
  "detail": "string"
}
```

5.5. Timestamp

Description: apply a timestamp to the submitted file.

URL: `https://{host}/signbox/api/v1/timestamp`

Method: POST

Header parameters:

- **No parameters**

Parameters in request body:

- **file**: required, represent the file to be timestamped
    - type: {STRING}
- **url**: required, represent the timestamp service URL

    - type: {STRING}
- **format**: supported timestamp formats
    - supported format:
        - PDF
        - P7M
    - type: {STRING}

Sample response in case of success:

- **HTML Code:** 200 OK
- **Content:** file timestamped

Sample response in case of error:

- **HTML Code:** 401 Unauthorized
- **Content:**

```
{
  "error_code": 0,
  "detail": "string"
}
```